Summary in Graph

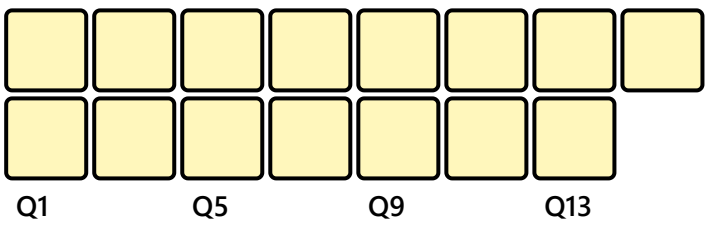
Exam Summary (GO Classes CS Test Series 2026 | Computer Networks | Topic Wise Test 3)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (102) out of 1022 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Computer Networks

A program on workstation A attempts to open a TCP connection to port 1FFD H on a machine B, and a UDP connection to port 5EFA H on machine C. Which of the two machines generate an ICMP port unreachable error if both the applications on the corresponding ports on B and C are down?

A. Only B
B. Only C

C. Both B and C
D. Neither B nor C

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 02sec | Discuss

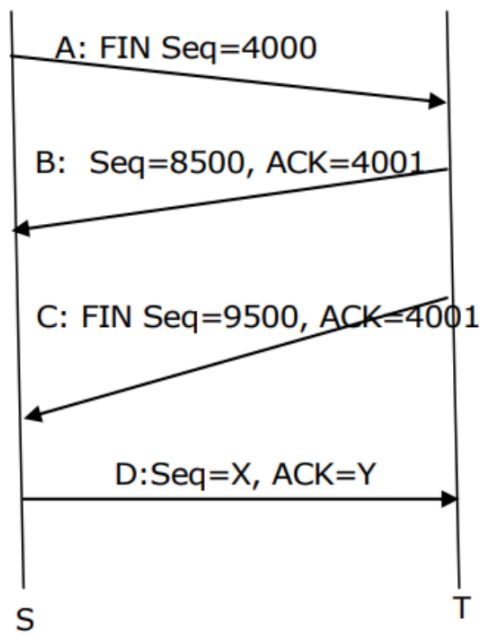
Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Computer Networks

Consider a TCP machine having window size 32 KB over a 512 Mbps channel that has one way delay of 20 msec. The line efficiency in percentage is _____

Your Answer: | Correct Answer: 1.26 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

TCP connection termination using the three-way handshake is shown below.



Consider the following statements about this termination:

I. Between B and C, two-way communication is still possible.

II. Between B and C, T can still send data but S cannot.

III. The value of X is 4000.

IV. The value of Y is 9500.

Assume FIN carries no data.

Which of the above statements is/are true?

A. II only
B. I, III, and IV only
C. I and II only
D. II, III, and IV only

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Computer Networks

Which of the following statements are true about TCP?

i. It is a byte-oriented port-to-port communication
ii. It uses flow control mechanism protocol
iii. Its connections are link to link and full duplex
iv. It uses piggybacking whenever possible

A. (i), (iii) and (iv) only
B. (i), (ii) and (iv) only

C. (ii), (iii) and (iv) only
D. All of the given

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Computer Networks

Which of the following statements regarding TCP are false?

A. A TCP transmitter normally interprets three duplicate ACKs to mean that, while data packets have been received out of order, all data is successfully being delivered.
B. TCP is a Transport layer protocol.
C. Acknowledging every TCP packet with a separate ACK is useful because it keeps the transmitters' congestion window size constant.
D. TCP is a network layer protocol.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

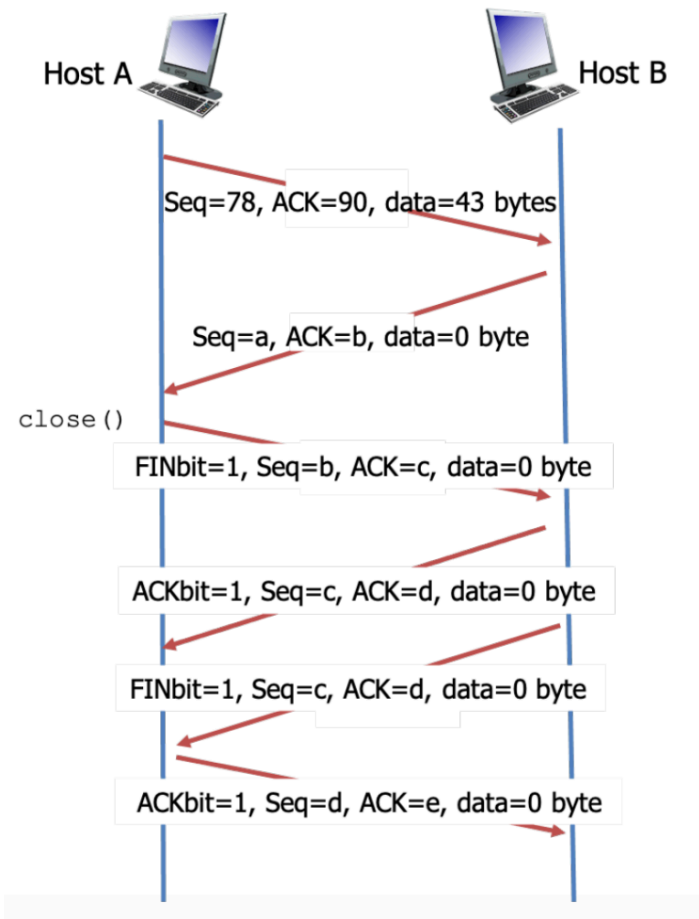
Host A has to 'inject' 30 Mbits of data into a network via a token bucket regulator. The token bucket has a capacity of 15 Mbits and is filled with tokens at the rate of 5 Mbps. Data is buffered if it arrives at the regulator when there are no tokens in the bucket. How long does it take, in total, for the 30 Mbits of data to enter the network, assuming that the host sends at a peak rate of 20 Mbps and the token bucket is initially full?

A. 3 seconds
B. 3.1 seconds
C. 4 seconds
D. 4.1 seconds

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

Depicted below is a sequence of packet exchange in a TCP connection between Host A and B. Host A sends the last data packet and receives the ACK packet from Host B. Then, Host A sends a FIN packet to initiate closing of the connection. Host B sends a FIN packet as well. Tell the value of a, b, c, d, e.



A. $a = c = 90$
B. $b = 121$
C. $e = 91$
D. $d = 121$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

A TCP connection starts with cwnd $= 1\text{MSS}$ and transmits packets of constant size 1MSS. Let's also assume the delayed ack is disabled for simplicity. With the initial cwnd $= 1\text{MSS}$, the sender sends the 1st data packet (sequence number 0) to the receiver. After receiving the 1st ACK packet (ack sequence number 1MSS) from the receiver, the sender increases the cwnd to 2MSS. The sender can therefore send the 2nd and 3rd data packet back-to-back. The 2nd and 3rd ACK packet will come back and trigger changes to cwnd subsequently. This will go on and all data and ACK packets will come through smoothly except for the 9th data packet. It is dropped. Listed below are the values of cwnd receiving 12 ACK packets respectively.

$$(2\text{MSS}, 3\text{MSS}, 4\text{MSS}, 4.25\text{MSS}, a, b, c, d, d, d, d, d, \ldots)$$

What is the value of a, b, c and d?
Fractional values in MSS can be converted to integer in bytes.

A. $a = 4.49\text{MSS}$
B. $b = 4.71\text{MSS}$
C. $c = 4.92\text{MSS}$
D. $d = 5.12\text{MSS}$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

A message consisting of 2400 Bytes is to be passed over the Internet. The message is passed to the transport layer which appends a 150-Bytes header, followed by the network layer which uses a 26-Byte header, followed by the Data link layer which uses a 26-Byte header. Packets are transmitted via two

networks (i.e., from the source host to a router to the destination host) with MTUs = 5000 Bytes and 900 Bytes respectively. The number of Bytes, including headers delivered to the destination network, is

A. 2706 Bytes
B. 2634 Bytes
C. 2554 Bytes
D. 2476 Bytes

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

Consider traces of TCP congestion window size recorded over time from 5 different connections.

- Connection $\mathrm{A}: 1, 2, 4, 8, 16, 17, 18, 19$
- Connection $\mathrm{B}: 1, 2, 4, 8, 9, 10, 11, 12$
- Connection $\mathrm{C}: 1, 2, 4, 5, 6, 3, 4, 5$
- Connection $\mathrm{D}: 1, 2, 3, 4, 1, 2, 3, 4$
- Connection $\mathrm{E}: 1, 2, 4, 8, 16, 32, 1, 2$

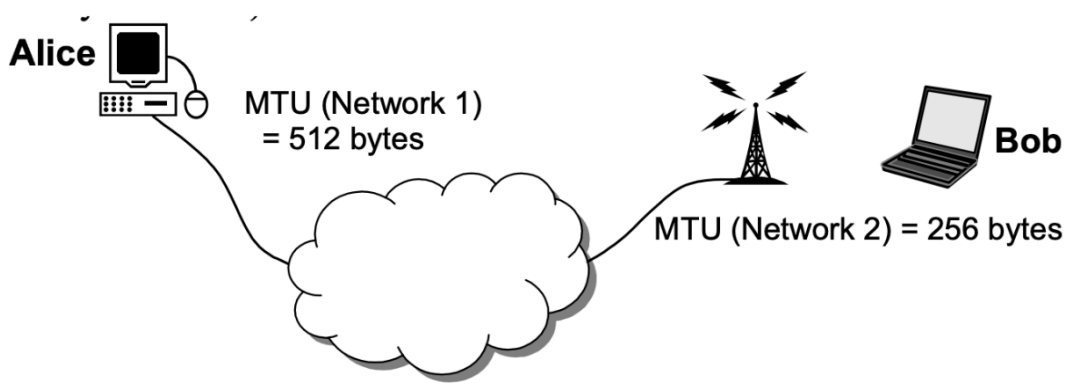
Which of the following is/are true about the above connections?

A. Except Connection E, all connections ended in the congestion avoidance state at the end of the recording
B. ONLY Connection C experienced a packet loss detected by receiving 3 duplicate acknowledgment packets
C. ONLY D experienced a packet loss detected by not receiving any new acknowledgment packet after a retransmission timeout event
D. ONLY E experienced a packet loss detected by not receiving any new acknowledgment packet after a retransmission timeout event

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Computer Networks

Alice is a musician and she wants to send her new song to Bob as a 2-MB $\mathrm{MP3}$ file. The bottleneck link between their computers has a 10-Mbps transmission rate. Assume the header overhead for link layer of both networks to be 26 bytes and all other headers used does not have options.



It is not known if TCP or UDP is used by sender. Which of the following is/are CORRECT ?

A. If UDP is used then 4386 datagrams are generated from Alice's computer.
B. If TCP is used then 4482 datagrams are generated from Alice.
C. If UDP is used then 13514 datagrams are recived at Bob's computer.
D. If TCP is used then 13444 datagrams are recived at Bob's computer.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

Assume that the TCP sequence number field is defined using 16 bits, so the sequence number space is $\{0, 1, 2, \ldots, 2^{16} - 1\}$, and the sequence number wraps around to 0 after reaching $2^{16} - 1$. A TCP connection has already been established, and the sender assigns the initial sequence number as $\mathrm{ISN} = 33535$. Assume that each increment of the TCP sequence number corresponds to exactly 1 byte of transmitted data, and the data transmission rate is $32\,\mathrm{MBps}$, where $1\,\mathrm{MB} = 10^6$ bytes. How long (in milliseconds) will it take for the TCP sequence number to wrap around and become 0 for the first time?

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

A TCP connection is established between two processes $\mathrm{P1}$ and $\mathrm{P2}$ running on different hosts. Segment size is set to 800 bytes for this connection. Initially, the ssthresh at $\mathrm{P1}$ is 7000 bytes. Then, $\mathrm{P1}$ receives a TCP ACK reporting a rwnd size of 4000 bytes. After reception and processing of this ACK, the cwnd value is 7200 bytes. How many maximum-sized segments can $\mathrm{P1}$ transmit now (i.e., after the ACK processing)?

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Computer Networks

The following is a dump of UDP header in hexadecimal format : 5EFA00FD001C3297.

What is the total length of user datagram?

A. 30 bytes
B. 28 bytes
C. 32 bytes
D. 24 bytes

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Numerical Type | Award: 2 | Penalty: 0 | Computer Networks

Assume that TCP implements an extension that allows window sizes much larger than $64\ \mathrm{KB}$. Suppose that you are using this extended TCP over a $1\text{-}\mathrm{Gbps}$ link with a constant one-way latency of $50\ \mathrm{ms}$ to transfer a $10\ \mathrm{MB}$ file, and the TCP receive window is $16\ \mathrm{MB}$. If TCP sends $1\ \mathrm{KB}$ packets (assuming no congestion and no lost packets). How many RTTs does it take to send the file?

Your Answer: | Correct Answer: 14 | Not Attempted | Time taken: 00min 00sec | Discuss

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